Dual Supergradient Ascent

We now develop a direct method for the dual problem studied in Chapters 18 and 19. The inner minimization makes the dual concave, but it may also create corners, even when every function in the primal formulation is differentiable. We will first describe first-order information at such a corner, obtain it from Lagrangian minimizers, and determine which directions actually increase the function. We will then construct a method using just one of those minimizers per iteration. Its analysis will explain why an algorithm can converge without increasing the dual value at every iteration, what a diminishing step accomplishes, and why an average of iterates is a useful output.

Learning objectives

By the end of this chapter, you should be able to:

- define a concave supergradient and compute superdifferentials of lower envelopes;
- obtain dual supergradients from global Lagrangian minimizers and state when the dual is differentiable;
- distinguish an arbitrary supergradient from a direction of steepest ascent;
- implement projected dual supergradient ascent and choose an appropriate step sequence;
- derive ergodic and best-value guarantees, and distinguish them from convergence of the iterates;
- relate these guarantees to convex subgradient descent; and
- check the additional requirements for inexact inner solves and primal recovery.

20.1 From a smooth primal problem to a nonsmooth dual

We retain the full-space formulation of the preceding two chapters:

$$\inf_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x}) \qquad \text{subject to} \qquad \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}, \quad \boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0}.$$

All functions are finite on $\mathbb{R}^n$, and every constraint is explicit. Define

$$\boldsymbol{w} = (\boldsymbol{\lambda}, \boldsymbol{\nu}), \qquad W = \mathbb{R}^m \times \mathbb{R}^p_+, \qquad \boldsymbol{r}(\boldsymbol{x}) = \begin{bmatrix} \boldsymbol{g}(\boldsymbol{x}) \\ \boldsymbol{h}(\boldsymbol{x}) \end{bmatrix}.$$

The Lagrangian and its dual function can then be written as

$$L(\boldsymbol{x}, \boldsymbol{w}) = f(\boldsymbol{x}) + \boldsymbol{w}^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}), \qquad \theta(\boldsymbol{w}) = \inf_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{w}), \qquad d^* = \sup_{\boldsymbol{w}\in W} \theta(\boldsymbol{w}). \tag{20.1}$$

For each fixed $\boldsymbol{x}$, the function $L(\boldsymbol{x}, \cdot)$ is affine. Its slope is the residual vector $\boldsymbol{r}(\boldsymbol{x})$. The dual function is their lower envelope: as the multiplier changes, the affine function attaining the lowest value can change. This is the source of nonsmoothness that we need to handle.

We construct an explicit envelope from a full-space problem, so that both its smooth pieces and its corners come from genuine Lagrangian inner minima. This illustration is separate from the common numerical test later in the chapter. Set

$$\phi(t) = 4\big(\max\{|t| - 1, 0\}\big)^2$$

and consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \ \phi(x_1) + \phi(x_2) - 2x_2 + \tfrac{1}{4} \qquad \text{subject to} \qquad r(\boldsymbol{x}) = x_1 + x_2 + \tfrac{3}{4} = 0. \tag{20.2}$$

The two coordinates separate in its Lagrangian. We therefore first minimize $\phi(t) + at$ for an arbitrary scalar a. For $a > 0$, the minimum lies to the left of $[-1, 1]$ and solves $8(t+1)+a = 0$. For $a < 0$, it lies to the right and solves $8(t-1)+a = 0$. When $a = 0$, every point in $[-1, 1]$ minimizes the expression. Substitution gives

$$\arg\min_t\{\phi(t) + at\} = \begin{cases} \{-\operatorname{sign}(a)(1 + |a|/8)\}, & a \neq 0, \\ [-1, 1], & a = 0, \end{cases} \qquad \inf_t\{\phi(t) + at\} = -|a| - \frac{a^2}{16}.$$

Using $a = w$ for the first coordinate and $a = w - 2$ for the second yields

$$\begin{aligned} \theta(w) &= \tfrac{1}{4} + \tfrac{3}{4}w - |w| - \frac{w^2}{16} - |w - 2| - \frac{(w-2)^2}{16} \\ &= -\frac{w^2}{8} + w - |w| - |w - 2|. \end{aligned} \tag{20.3}$$

This function is finite for every w. Its inner minimizer is unique away from $w = 0, 2$; at either of those values one coordinate ranges over a flat interval. The corresponding affine functions have different slopes and meet at the corners marked A and B in Figure 20.1. We will use these same corners to study ascent and optimality.

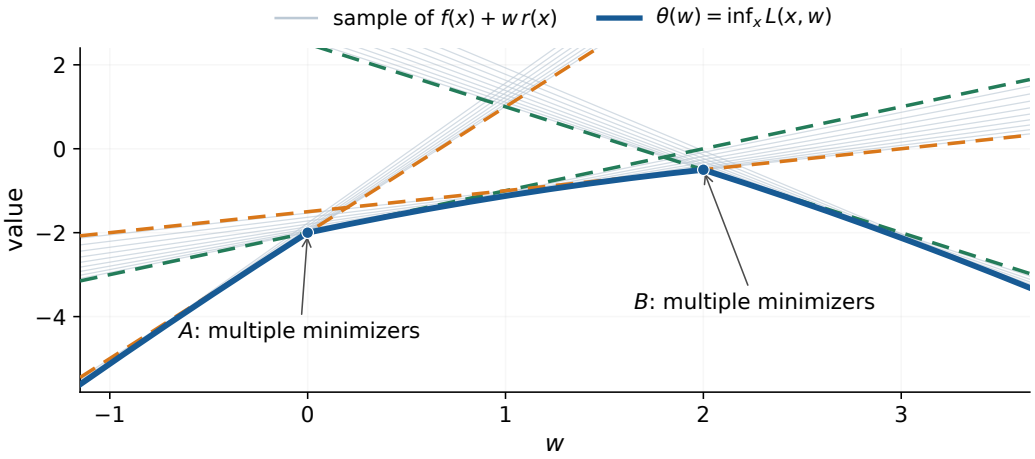


Figure 20.1: The dual is the lower envelope of infinitely many affine functions $f(\boldsymbol{x}) + wr(\boldsymbol{x})$. Pale lines show a finite sample; the dark curve is the exact envelope. The colored dashed lines attain the inner minimum at A or B. Their distinct slopes produce the two corners.

20.2 Supergradients and their geometry

The simplest illustration is

$$q(w) = \min\{w, -w\} = -|w|.$$

For $w < 0$ the active line has slope 1, while for $w > 0$ it has slope -1. At zero both lines are active. There is no single tangent line whose slope represents the two one-sided changes. Nevertheless, every line through the origin with slope between -1 and 1 lies above the graph. This suggests replacing the tangent gradient by the slopes of all supporting affine functions.

Definition 20.1 (Supergradient and superdifferential)**.** Let $\theta : \mathbb{R}^q \to \mathbb{R} \cup \{-\infty\}$ be concave, and let $\theta(\boldsymbol{w})$ be finite. A vector $\boldsymbol{\xi} \in \mathbb{R}^q$ is a *supergradient* of θ at $\boldsymbol{w}$ if

$$\theta(\boldsymbol{z}) \le \theta(\boldsymbol{w}) + \boldsymbol{\xi}^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{w}) \qquad \text{for every } \boldsymbol{z} \in \mathbb{R}^q. \tag{20.4}$$

The set of all such vectors, denoted $\partial^{+}\theta(\boldsymbol{w})$, is the *superdifferential*.

The inequality means that the affine function on its right supports the graph from above and agrees with it at $\boldsymbol{w}$. The superdifferential is a closed convex set, since each choice of $\boldsymbol{z}$ imposes a closed linear inequality on $\boldsymbol{\xi}$. It is not necessarily nonempty at a boundary point of the effective domain $\operatorname{dom}\theta = \{\boldsymbol{w} : \theta(\boldsymbol{w}) > -\infty\}$. At interior points where a concave function is finite, supporting hyperplanes do exist.

To compare with the usual convex terminology, put $F = -\theta$ and multiply Equation (20.4) by -1. This gives

$$\boldsymbol{\xi} \in \partial^{+}\theta(\boldsymbol{w}) \quad \Longleftrightarrow \quad -\boldsymbol{\xi} \in \partial F(\boldsymbol{w}). \tag{20.5}$$

Thus *supergradient ascent* for a concave function is precisely *subgradient descent* for its negative. This explains the conventional name *dual subgradient method*; the vectors used for ascent in this chapter are called supergradients to keep their sign explicit.

For $q(w) = -|w|$, direct substitution into the definition gives

$$\partial^{+}q(w) = \begin{cases} \{1\}, & w < 0, \\ [-1, 1], & w = 0, \\ \{-1\}, & w > 0. \end{cases}$$

At a differentiable interior point, the only supergradient is the ordinary gradient: apply Equation (20.4) to $\boldsymbol{z} = \boldsymbol{w} + t\boldsymbol{d}$, divide by $t > 0$, and let $t \downarrow 0$. Repeating with $-\boldsymbol{d}$ forces $\boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{d} = \nabla\theta(\boldsymbol{w})^{\mathsf{T}}\boldsymbol{d}$ for every $\boldsymbol{d}$.

20.2.1 Computing first-order information from the inner problem

To apply the definition to a dual function, we need to obtain a supporting slope without differentiating the optimal primal point. Let

$$X(\boldsymbol{w}) = \arg\min_{\boldsymbol{x} \in \mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{w}).$$

First suppose this set is nonempty, and select one $\boldsymbol{x}(\boldsymbol{w}) \in X(\boldsymbol{w})$. Holding this primal point fixed, evaluate the Lagrangian at a different multiplier:

$$\theta(\boldsymbol{z}) \le L(\boldsymbol{x}(\boldsymbol{w}), \boldsymbol{z}) = L(\boldsymbol{x}(\boldsymbol{w}), \boldsymbol{w}) + \boldsymbol{r}(\boldsymbol{x}(\boldsymbol{w}))^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{w}).$$

Since $L(\boldsymbol{x}(\boldsymbol{w}), \boldsymbol{w}) = \theta(\boldsymbol{w})$, this is the required supporting inequality. We record the result because it is the computational basis of the method.

Lemma 20.1 (Residual supergradient)**.** *Every global inner minimizer* $\boldsymbol{x}(\boldsymbol{w}) \in X(\boldsymbol{w})$

produces a dual supergradient:

$$\boldsymbol{r}(\boldsymbol{x}(\boldsymbol{w})) \in \partial^{+}\theta(\boldsymbol{w}).$$

Neither convexity of the primal problem nor uniqueness of the inner minimizer is needed for this statement.

The global qualification is essential. A local minimizer or a point with a small stationarity residual need not attain the infimum. In that case its Lagrangian value is an upper estimate of $\theta(\boldsymbol{w})$, and the displayed argument no longer gives an exact supporting affine function.

20.2.2 Directional derivatives at a corner

One inner minimizer gives one supporting slope. To determine the whole set of slopes and relate it to differentiability, we first establish the local geometry needed in the next subsection. For a direction $\boldsymbol{d}$, define the one-sided directional derivative

$$\theta'(\boldsymbol{w};\boldsymbol{d}) = \lim_{t\downarrow 0} \frac{\theta(\boldsymbol{w}+t\boldsymbol{d}) - \theta(\boldsymbol{w})}{t}.$$

When θ is finite in a neighborhood of $\boldsymbol{w}$, concavity ensures that the limit is finite. The difference quotients are nonincreasing as t increases, and local boundedness of their slopes follows from the values of θ on a slightly larger ball. The derivative may fail to be linear in $\boldsymbol{d}$.

Lemma 20.2 (Directional information and differentiability)**.** *Suppose θ is concave and finite on an open neighborhood of $\boldsymbol{w}$. The nearby superdifferentials are nonempty and locally uniformly bounded. In particular, $\partial^{+}\theta(\boldsymbol{w})$ is nonempty and compact, and*

$$\theta'(\boldsymbol{w};\boldsymbol{d}) = \min_{\boldsymbol{\xi}\in\partial^{+}\theta(\boldsymbol{w})} \boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{d}. \tag{20.6}$$

Moreover, θ is differentiable at $\boldsymbol{w}$ if and only if $\partial^{+}\theta(\boldsymbol{w})$ is a singleton.

Proof. We use the supporting-hyperplane principle for a convex set, as in Chapter 19. A finite concave function is locally continuous and locally Lipschitz in the interior of its domain. One way to see the latter fact is to bound its values above and below on a ball and apply the one-dimensional concavity inequalities along chords of a larger ball: the oscillation of the function divided by the distance to the larger ball's boundary bounds every interior chord slope. The hypograph is convex, so a supporting hyperplane at $(\boldsymbol{w},\theta(\boldsymbol{w}))$ gives a vector satisfying Equation (20.4). The coefficient of the vertical coordinate must be positive: it cannot be negative because the hypograph extends downward, and it cannot be zero because $\boldsymbol{w}$ is an interior point of the domain. Normalizing that coefficient gives a supergradient. A local supporting inequality extends to the entire effective domain by concavity along segments.

We next need a bound that holds at all nearby points, because the reverse directional inequality will use supergradients at $\boldsymbol{w}+t\boldsymbol{d}$, not just at $\boldsymbol{w}$. Choose $\delta > 0$ so that the closed ball $\overline{B}(\boldsymbol{w},2\delta)$ lies in the open neighborhood where θ is finite, and put

$$M = \max_{\boldsymbol{y}\in\overline{B}(\boldsymbol{w},2\delta)} |\theta(\boldsymbol{y})| < \infty.$$

Continuity gives this finite bound. The preceding supporting-hyperplane argument also gives a supergradient at every point of the smaller ball. For any $\boldsymbol{z} \in B(\boldsymbol{w},\delta)$, any $\boldsymbol{\xi} \in \partial^{+}\theta(\boldsymbol{z})$,

and any unit vector $\boldsymbol{u}$, the points $\boldsymbol{z} \pm \delta \boldsymbol{u}$ lie in the larger ball. Evaluating the supporting inequality at these two points gives

$$\frac{\theta(\boldsymbol{z}+\delta\boldsymbol{u})-\theta(\boldsymbol{z})}{\delta} \le \boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{u} \le \frac{\theta(\boldsymbol{z})-\theta(\boldsymbol{z}-\delta\boldsymbol{u})}{\delta}.$$

Both outer expressions have absolute value at most $2M/\delta$. If $\boldsymbol{\xi} \neq \boldsymbol{0}$, take $\boldsymbol{u} = \boldsymbol{\xi}/\|\boldsymbol{\xi}\|$; the zero vector satisfies the same bound. We have therefore proved

$$\|\boldsymbol{\xi}\| \le \frac{2M}{\delta} \qquad \text{for every } \boldsymbol{z} \in B(\boldsymbol{w},\delta), \quad \boldsymbol{\xi} \in \partial^{+}\theta(\boldsymbol{z}). \tag{20.7}$$

This is *local uniform boundedness*: one constant controls all the nearby superdifferentials. Closedness then makes $\partial^{+}\theta(\boldsymbol{w})$ compact.

For any $\boldsymbol{\xi} \in \partial^{+}\theta(\boldsymbol{w})$, the supporting inequality gives $\theta'(\boldsymbol{w};\boldsymbol{d}) \le \boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{d}$. To prove the reverse inequality, choose a sequence $t_j \downarrow 0$ with $\boldsymbol{z}_j = \boldsymbol{w} + t_j\boldsymbol{d} \in B(\boldsymbol{w},\delta)$, and select any $\boldsymbol{\xi}_j \in \partial^{+}\theta(\boldsymbol{z}_j)$. Its supporting inequality evaluated at $\boldsymbol{w}$ gives

$$\frac{\theta(\boldsymbol{w}+t_j\boldsymbol{d})-\theta(\boldsymbol{w})}{t_j} \ge \boldsymbol{\xi}_j^{\mathsf{T}}\boldsymbol{d}.$$

The uniform bound in Equation (20.7) supplies a subsequence $\boldsymbol{\xi}_{j_\ell} \to \overline{\boldsymbol{\xi}}$. To identify its limit, fix any $\boldsymbol{y}$ in the effective domain and write

$$\theta(\boldsymbol{y}) \le \theta(\boldsymbol{z}_{j_\ell}) + \boldsymbol{\xi}_{j_\ell}^{\mathsf{T}}(\boldsymbol{y}-\boldsymbol{z}_{j_\ell}).$$

Since $\boldsymbol{z}_{j_\ell} \to \boldsymbol{w}$ and θ is continuous at $\boldsymbol{w}$, passing to the limit gives the supporting inequality at $\boldsymbol{w}$ for every $\boldsymbol{y}$. Thus $\overline{\boldsymbol{\xi}} \in \partial^{+}\theta(\boldsymbol{w})$, and

$$\theta'(\boldsymbol{w};\boldsymbol{d}) = \lim_{\ell\to\infty} \frac{\theta(\boldsymbol{w}+t_{j_\ell}\boldsymbol{d})-\theta(\boldsymbol{w})}{t_{j_\ell}} \ge \overline{\boldsymbol{\xi}}^{\mathsf{T}}\boldsymbol{d} \ge \min_{\boldsymbol{\xi}\in\partial^{+}\theta(\boldsymbol{w})} \boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{d}.$$

Together with the upper inequality, this proves Equation (20.6). Only a convergent *subsequence* is needed; arbitrary choices of nearby supergradients need not converge as a whole. Boundedness of the single set $\partial^{+}\theta(\boldsymbol{w})$ is not the bound used to extract this subsequence.

Differentiability implies a singleton by the calculation preceding the lemma. For the converse, suppose $\partial^{+}\theta(\boldsymbol{w}) = \{\boldsymbol{\xi}\}$. For a small vector $\boldsymbol{u}$, choose $\boldsymbol{\xi}_{\boldsymbol{u}} \in \partial^{+}\theta(\boldsymbol{w}+\boldsymbol{u})$. The same uniform bound and limiting supporting inequality now show $\boldsymbol{\xi}_{\boldsymbol{u}} \to \boldsymbol{\xi}$ as $\boldsymbol{u} \to \boldsymbol{0}$. Indeed, otherwise there would be a sequence $\boldsymbol{u}_j \to \boldsymbol{0}$ whose selected supergradients stay a fixed positive distance from $\boldsymbol{\xi}$. A convergent subsequence would have its limit in $\partial^{+}\theta(\boldsymbol{w}) = \{\boldsymbol{\xi}\}$, a contradiction. The singleton assumption is what upgrades subsequential convergence to full convergence here. The two supporting inequalities now give

$$\boldsymbol{\xi}_{\boldsymbol{u}}^{\mathsf{T}}\boldsymbol{u} \le \theta(\boldsymbol{w}+\boldsymbol{u})-\theta(\boldsymbol{w}) \le \boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{u}.$$

The error after subtracting $\boldsymbol{\xi}^{\mathsf{T}}\boldsymbol{u}$ is at most $\|\boldsymbol{u}\|\,\|\boldsymbol{\xi}_{\boldsymbol{u}}-\boldsymbol{\xi}\| = o(\|\boldsymbol{u}\|)$. This is differentiability with gradient $\boldsymbol{\xi}$. □

20.2.3 All active residuals and Danskin's formula

One minimizer gives one supergradient. To describe the complete superdifferential, we need a condition preventing minimizers from escaping to infinity as the multiplier changes. This replaces the compact retained-domain assumption often used to state Danskin's theorem, while keeping our inner minimization over all of $\mathbb{R}^n$.

Theorem 20.3 (Local Danskin formula for the dual function)**.** *Suppose f and $\boldsymbol{r}$ are continuous. At a multiplier $\boldsymbol{w}$, assume there exist an open neighborhood U and a compact set $K \subset \mathbb{R}^n$ such that*

$$\varnothing \neq X(\boldsymbol{z}) \subset K \qquad (\boldsymbol{z} \in U).$$

Then θ is finite near $\boldsymbol{w}$, and

$$\theta'(\boldsymbol{w}; \boldsymbol{d}) = \min_{\boldsymbol{x} \in X(\boldsymbol{w})} \boldsymbol{d}^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}), \tag{20.8}$$

$$\partial^{+}\theta(\boldsymbol{w}) = \operatorname{conv}\{\boldsymbol{r}(\boldsymbol{x}) : \boldsymbol{x} \in X(\boldsymbol{w})\}. \tag{20.9}$$

Here conv *denotes the convex hull.*

Proof. For $\boldsymbol{z} \in U$, the full-space infimum equals the minimum over K. Continuity therefore makes $X(\boldsymbol{z})$ nonempty and compact and makes the value function continuous on U. More explicitly, a fixed current minimizer gives the upper limit inequality for the values; for the lower limit, select nearby minimizers in K, pass to a convergent subsequence, and use continuity.

For $\boldsymbol{x} \in X(\boldsymbol{w})$ and small $t > 0$,

$$\theta(\boldsymbol{w} + t\boldsymbol{d}) \leq L(\boldsymbol{x}, \boldsymbol{w} + t\boldsymbol{d}) = \theta(\boldsymbol{w}) + t\boldsymbol{d}^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}).$$

Taking the smallest right-hand slope proves the upper bound in Equation (20.8). For the lower bound, choose $\boldsymbol{x}_t \in X(\boldsymbol{w} + t\boldsymbol{d})$. Then

$$\frac{\theta(\boldsymbol{w} + t\boldsymbol{d}) - \theta(\boldsymbol{w})}{t} = \frac{L(\boldsymbol{x}_t, \boldsymbol{w}) - \theta(\boldsymbol{w})}{t} + \boldsymbol{d}^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}_t) \geq \boldsymbol{d}^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}_t).$$

Every sequence $t_j \downarrow 0$ has a subsequence for which $\boldsymbol{x}_{t_j} \to \overline{\boldsymbol{x}} \in K$. Passing to the limit in the optimality inequality shows $\overline{\boldsymbol{x}} \in X(\boldsymbol{w})$. The lower limit is therefore at least the minimum of $\boldsymbol{d}^{\mathsf{T}} \boldsymbol{r}$ on $X(\boldsymbol{w})$, proving the derivative formula.

Every active residual is a supergradient by Lemma 20.1; convexity of the superdifferential gives one inclusion in Equation (20.9). The residual image of $X(\boldsymbol{w})$ is compact, and its finite-dimensional convex hull is compact. If a supergradient $\boldsymbol{\xi}$ lay outside that hull, strict separation would give some $\boldsymbol{d}$ with

$$\boldsymbol{\xi}^{\mathsf{T}} \boldsymbol{d} < \min_{\boldsymbol{x} \in X(\boldsymbol{w})} \boldsymbol{r}(\boldsymbol{x})^{\mathsf{T}} \boldsymbol{d}.$$

But Equation (20.4) implies $\theta'(\boldsymbol{w}; \boldsymbol{d}) \leq \boldsymbol{\xi}^{\mathsf{T}} \boldsymbol{d}$, contradicting the derivative formula. This proves the reverse inclusion. □

A useful way to verify the compactness hypothesis is local uniform coercivity: if $L(\boldsymbol{x}, \boldsymbol{z}) \to +\infty$ as $\|\boldsymbol{x}\| \to \infty$ uniformly for $\boldsymbol{z}$ in a neighborhood, then nearby minimizers are attained and lie in one compact set. Continuity also makes their residuals uniformly bounded there. These are local conclusions; a convergence theorem may require a bound valid along an entire infinite sequence.

Combining Theorem 20.3 with Lemma 20.2 gives the exact smoothness statement:

$$\theta \text{ is differentiable at } \boldsymbol{w} \iff \{\boldsymbol{r}(\boldsymbol{x}) : \boldsymbol{x} \in X(\boldsymbol{w})\} \text{ is a singleton.} \tag{20.10}$$

In particular, a unique inner minimizer is sufficient, and its residual is $\nabla\theta(\boldsymbol{w})$. The converse is false: different minimizers can have the same residual. For example, take

$$f(x_1, x_2) = \tfrac{1}{2}x_1^2 + (x_2^2 - 1)^2, \qquad g(x_1, x_2) = x_1.$$

Then $X(\lambda) = \{(-\lambda, 1), (-\lambda, -1)\}$ for every λ, while $\theta(\lambda) = -\lambda^2/2$ is smooth. Both minimizers have the same equality residual $-\lambda$.

To apply the same reasoning without solving an inner optimization problem, consider a finite lower envelope $\theta(\boldsymbol{w}) = \min_i\{a_i + \boldsymbol{b}_i^\mathsf{T}\boldsymbol{w}\}$, the same reasoning gives

$$\partial^+\theta(\boldsymbol{w}) = \operatorname{conv}\{\boldsymbol{b}_i : a_i + \boldsymbol{b}_i^\mathsf{T}\boldsymbol{w} = \theta(\boldsymbol{w})\}.$$

Only the active affine pieces contribute. This is also a convenient way to compute the subdifferential of a convex maximum of affine functions after a sign change.

20.3 Which supergradient should be used?

20.3.1 A supergradient can point downhill

The supporting inequality bounds the function *above*, so a positive inner product with one supergradient does not guarantee improvement. At the maximum of $q(w) = -|w|$, the vector 1 belongs to $\partial^+ q(0)$, yet every positive step decreases q. This failure is not confined to an optimizer. Consider

$$q(w_1, w_2) = \min\{w_1 + 2w_2,\ w_1 - 2w_2\} = w_1 - 2|w_2|.$$

At the origin,

$$\partial^+ q(\mathbf{0}) = \operatorname{conv}\{(1, 2)^\mathsf{T}, (1, -2)^\mathsf{T}\}.$$

Choosing $\boldsymbol{\xi} = (1, 2)^\mathsf{T}$ gives $q(t\boldsymbol{\xi}) = -3t < q(\mathbf{0})$ for every $t > 0$, although moving in the horizontal direction $(1, 0)^\mathsf{T}$ increases the function.

To identify an ascent direction correctly, use the directional formula:

$$\theta'(\boldsymbol{w}; \boldsymbol{d}) = \min_{\boldsymbol{\xi} \in \partial^+\theta(\boldsymbol{w})} \boldsymbol{\xi}^\mathsf{T}\boldsymbol{d}.$$

Thus strict first-order ascent requires a positive inner product with *every* supporting slope. For a concave function finite near $\boldsymbol{w}$, this is also equivalent to actual local strict ascent. Indeed, a positive derivative makes sufficiently small difference quotients positive. Conversely, if $\theta(\boldsymbol{w} + t_0\boldsymbol{d}) > \theta(\boldsymbol{w})$ for any $t_0 > 0$, concavity makes the right derivative at zero at least that positive secant slope.

20.3.2 The direction of steepest ascent

We now maximize the instantaneous increase under a unit-length constraint:

$$\max_{\|\boldsymbol{d}\| \le 1} \theta'(\boldsymbol{w}; \boldsymbol{d}).$$

The inequality $\|\boldsymbol{d}\| \le 1$ includes $\boldsymbol{d} = \mathbf{0}$, which is needed when no direction improves the function.

Theorem 20.4 (Minimum-norm supergradient and steepest ascent)**.** *Suppose θ is concave and finite in a neighborhood of $\boldsymbol{w}$, and let*

$$\widehat{\boldsymbol{\xi}} = \operatorname*{arg\,min}_{\boldsymbol{\xi} \in \partial^+\theta(\boldsymbol{w})} \|\boldsymbol{\xi}\|.$$

Then a direction of steepest ascent is

$$\widehat{\boldsymbol{d}} = \begin{cases} \widehat{\boldsymbol{\xi}} / \left\|\widehat{\boldsymbol{\xi}}\right\|, & \widehat{\boldsymbol{\xi}} \neq \mathbf{0}, \\ \mathbf{0}, & \widehat{\boldsymbol{\xi}} = \mathbf{0}, \end{cases} \qquad \max_{\|\boldsymbol{d}\| \leq 1} \theta'(\boldsymbol{w}; \boldsymbol{d}) = \left\|\widehat{\boldsymbol{\xi}}\right\|.$$

Proof. The closed convex superdifferential has a unique minimum-norm point, since the squared norm is strictly convex. For any $\boldsymbol{\xi}$ in that set, the segment from $\widehat{\boldsymbol{\xi}}$ to $\boldsymbol{\xi}$ stays in the set. Differentiating the squared norm along the segment at its minimum gives

$$\widehat{\boldsymbol{\xi}}^{\mathsf{T}}(\boldsymbol{\xi} - \widehat{\boldsymbol{\xi}}) \geq 0.$$

For any $\|\boldsymbol{d}\| \leq 1$,

$$\theta'(\boldsymbol{w}; \boldsymbol{d}) = \min_{\boldsymbol{\xi} \in \partial^+ \theta(\boldsymbol{w})} \boldsymbol{\xi}^{\mathsf{T}} \boldsymbol{d} \leq \widehat{\boldsymbol{\xi}}^{\mathsf{T}} \boldsymbol{d} \leq \left\|\widehat{\boldsymbol{\xi}}\right\|.$$

If $\widehat{\boldsymbol{\xi}} \neq \mathbf{0}$, use the segment inequality to obtain

$$\min_{\boldsymbol{\xi} \in \partial^+ \theta(\boldsymbol{w})} \frac{\boldsymbol{\xi}^{\mathsf{T}} \widehat{\boldsymbol{\xi}}}{\left\|\widehat{\boldsymbol{\xi}}\right\|} = \left\|\widehat{\boldsymbol{\xi}}\right\|,$$

with equality attained at $\boldsymbol{\xi} = \widehat{\boldsymbol{\xi}}$. The normalized minimum-norm vector attains the upper bound. If it is zero, every directional derivative is nonpositive and the zero direction attains zero. □

In the two-dimensional example above, the closest point of the vertical segment to the origin is $(1, 0)^{\mathsf{T}}$, so the horizontal direction is steepest.

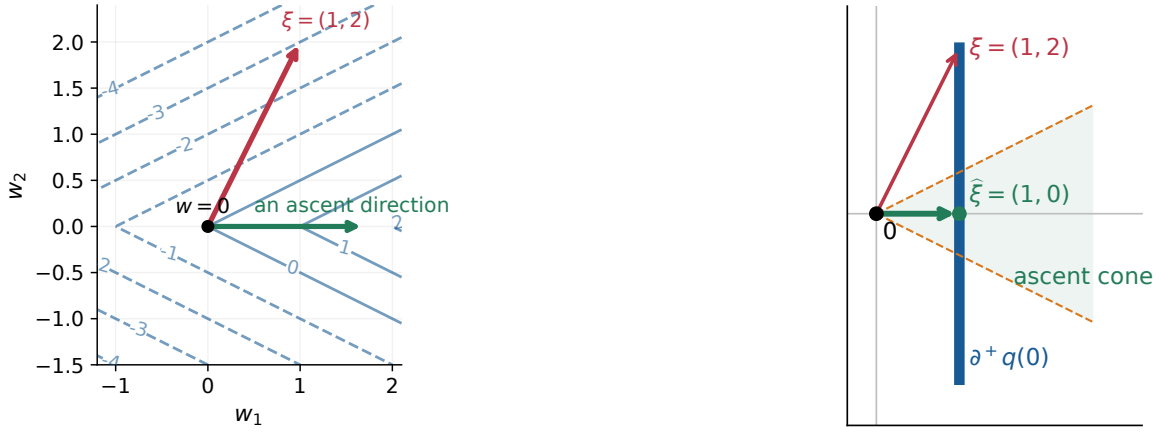


Figure 20.2: Left: an active supporting slope can point downhill. Right: the minimum-norm supergradient selects the steepest ascent direction.

For convex minimization, the corresponding steepest descent direction is the *negative* of the normalized minimum-norm subgradient. The sign changes because the objective is negated.

This theorem concerns unconstrained local directions. At a boundary point of W, one must restrict the direction to the tangent cone of W. The unconstrained minimum-norm formula does not automatically solve that different problem. In particular, projecting a chosen supergradient does not turn it into a steepest feasible ascent direction.

20.3.3 An ascending corner and an optimal corner

Return to the envelope in Equation (20.3). At A, where $w = 0$, the scalar inner solutions give

$$X(0) = [-1, 1] \times \{5/4\}, \qquad \{r(\boldsymbol{x}) : \boldsymbol{x} \in X(0)\} = [1, 3].$$

These minimizers remain in a common compact set when w varies in a bounded neighborhood, so Danskin's formula applies. Hence $\partial^+\theta(0) = [1, 3]$. The directional formula now gives

$$\theta'(0; +1) = 1, \qquad \theta'(0; -1) = -3.$$

The minimum-norm supergradient is 1, so the steepest unit direction is $+1$ and its slope is 1. In particular, this corner is not a maximum: the lower envelope can still rise to the right. Its rightward slope is the smaller active slope, even though a supporting line of slope 3 also passes through the point.

At B, where $w = 2$, the corresponding calculation gives

$$X(2) = \{-5/4\} \times [-1, 1], \qquad \partial^+\theta(2) = [-3/2, 1/2].$$

This time the closest supergradient to zero is zero itself. To interpret that fact, substitute the zero vector into the supporting inequality. It gives the unconstrained optimality test

$$\mathbf{0} \in \partial^+\theta(\boldsymbol{w}^*) \quad \Longleftrightarrow \quad \theta(\boldsymbol{w}^*) \geq \theta(\boldsymbol{w}) \text{ for every } \boldsymbol{w}. \tag{20.11}$$

At a constrained maximizer over W, zero need not be a supergradient: a boundary can block all improving directions. We will account for that boundary by projection rather than applying Equation (20.11) unchanged.

For this envelope, the horizontal support at B gives $\theta(w) \leq \theta(2) = -1/2$ for every w. The primal point $\boldsymbol{x} = (-5/4, 1/2)$ is an inner minimizer at $w = 2$ and satisfies $r(\boldsymbol{x}) = 0$. Consequently its objective value equals $\theta(2)$, providing a matching primal–dual certificate. At the simpler corner $q(w) = -|w|$, both unit directional derivatives are $q'(0; 1) = q'(0; -1) = -1$; its superdifferential $[-1, 1]$ likewise contains zero.

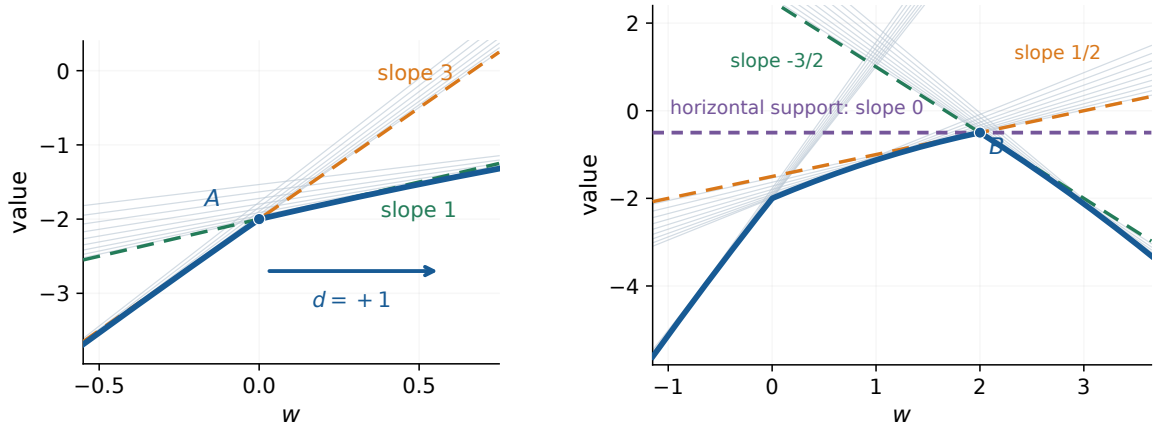


Figure 20.3: At A, every supporting slope is positive and the envelope increases to the right. At B, a horizontal supporting line certifies a global maximum. A corner alone does not determine whether ascent is possible.

20.3.4 Why the basic algorithm uses one residual instead

To implement the steepest rule, we need enough information about the whole superdifferential to locate its minimum-norm point. By Danskin's formula, finding it may require all global inner minimizers and a further minimum-norm problem over their residuals. For a known finite list of active residuals $\boldsymbol{r}_1, \dots, \boldsymbol{r}_s$, that extra problem is

$$\min_{\gamma_i \geq 0,\, \sum_i \gamma_i = 1} \frac{1}{2} \left\| \sum_i \gamma_i \boldsymbol{r}_i \right\|^2 .$$

Obtaining the list can already be much harder than computing a single inner solution. The basic dual method therefore solves one inner problem, uses its residual, and accepts that the next dual value may be smaller. Any exact inner minimizer is allowed, including a deterministic choice when there is a tie. Convergence will follow from the aggregate effect of the steps, rather than from monotone improvement.

This distinction resolves two different statements. An arbitrary supergradient is not necessarily an ascent direction, and a constant-step method using it need not converge. It does *not* follow that arbitrary supergradient choices invalidate all convergence guarantees: bounded exact choices with suitable diminishing steps satisfy the results below.

20.4 The projected dual supergradient algorithm

20.4.1 Checking the inner problem and the projection set

Projection onto W enforces only $\boldsymbol{\nu} \geq \mathbf{0}$. It does not ensure $\theta(\boldsymbol{w}) > -\infty$, and it does not make an unattained inner infimum attained. For example, the primal problem $\min_x x$ subject to $-x \leq 0$ has the dual function

$$\theta(\nu) = \inf_{x \in \mathbb{R}} \{x + \nu(-x)\} = \begin{cases} 0, & \nu = 1, \\ -\infty, & \nu \neq 1. \end{cases}$$

The effective domain is only $\{1\}$, although every $\nu \geq 0$ satisfies the multiplier sign restriction. At $\nu = 1$, every x is an inner minimizer; an arbitrary residual step can immediately leave the effective domain. Thus the algorithm requires valid inner solves at every generated point. If necessary, one may instead project onto a known nonempty closed convex set $C \subset W \cap \operatorname{dom} \theta$ on which the oracle is valid. To solve the original dual, that set must contain a dual optimizer; otherwise one is solving a restricted dual problem. The convergence argument below applies to either choice of C.

20.4.2 Projection and multiplier updates

For a nonempty closed convex set C, let

$$\Pi_C(\boldsymbol{z}) = \operatorname*{arg\,min}_{\boldsymbol{y} \in C} \frac{1}{2} \|\boldsymbol{y} - \boldsymbol{z}\|^2 .$$

The minimizer exists by closedness and coercivity and is unique by strict convexity. Projection enforces membership of the multiplier set after an otherwise unrestricted step.

Lemma 20.5 (Projection inequality and nonexpansiveness)**.** *For every $\boldsymbol{z} \in \mathbb{R}^q$ and $\boldsymbol{y} \in C$,*

$$(\boldsymbol{z} - \Pi_C(\boldsymbol{z}))^{\mathsf{T}}(\boldsymbol{y} - \Pi_C(\boldsymbol{z})) \le 0. \tag{20.12}$$

Consequently,

$$\|\Pi_C(\boldsymbol{z}_1) - \Pi_C(\boldsymbol{z}_2)\| \le \|\boldsymbol{z}_1 - \boldsymbol{z}_2\|. \tag{20.13}$$

Proof. Write $\boldsymbol{p} = \Pi_C(\boldsymbol{z})$. The segment $\boldsymbol{p} + t(\boldsymbol{y} - \boldsymbol{p})$ is feasible for $0 \le t \le 1$. The right derivative of its squared distance to $\boldsymbol{z}$ is nonnegative at $t = 0$, which gives $(\boldsymbol{p}-\boldsymbol{z})^{\mathsf{T}}(\boldsymbol{y}-\boldsymbol{p}) \ge 0$. For nonexpansiveness, put $\boldsymbol{p}_i = \Pi_C(\boldsymbol{z}_i)$ and apply this inequality with $(\boldsymbol{z}, \boldsymbol{y}) = (\boldsymbol{z}_1, \boldsymbol{p}_2)$ and $(\boldsymbol{z}_2, \boldsymbol{p}_1)$. Adding yields

$$\|\boldsymbol{p}_1 - \boldsymbol{p}_2\|^2 \le (\boldsymbol{p}_1 - \boldsymbol{p}_2)^{\mathsf{T}}(\boldsymbol{z}_1 - \boldsymbol{z}_2) \le \|\boldsymbol{p}_1 - \boldsymbol{p}_2\| \, \|\boldsymbol{z}_1 - \boldsymbol{z}_2\|.$$

Divide by $\|\boldsymbol{p}_1 - \boldsymbol{p}_2\|$ when it is nonzero; otherwise the claim is immediate. □

Starting from $\boldsymbol{w}^{(0)} \in W$, choose one global inner minimizer and set

$$\boldsymbol{x}^{(k)} \in X(\boldsymbol{w}^{(k)}), \qquad \boldsymbol{\xi}^{(k)} = \boldsymbol{r}(\boldsymbol{x}^{(k)}), \qquad \boldsymbol{w}^{(k+1)} = \Pi_W(\boldsymbol{w}^{(k)} + \alpha^{(k)}\boldsymbol{\xi}^{(k)}). \tag{20.14}$$

Here $\alpha^{(k)} > 0$ is a step coefficient; the actual unprojected displacement has length $\alpha^{(k)} \left\|\boldsymbol{\xi}^{(k)}\right\|$. Since W is a product of free equality multipliers and a nonnegative orthant, projection gives

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha^{(k)}\boldsymbol{g}(\boldsymbol{x}^{(k)}), \tag{20.15}$$

$$\boldsymbol{\nu}^{(k+1)} = \left[\boldsymbol{\nu}^{(k)} + \alpha^{(k)}\boldsymbol{h}(\boldsymbol{x}^{(k)})\right]_+. \tag{20.16}$$

A violated inequality increases its multiplier. A slack inequality decreases its multiplier, with projection preventing it from becoming negative.

Algorithm: Projected dual supergradient ascent

Choose $\boldsymbol{w}^{(0)} \in W \cap \operatorname{dom}\theta$ with an attained inner minimum, a positive step sequence, and an iteration budget K. For $k = 0, \dots, K-1$:

1. Compute a global minimizer $\boldsymbol{x}^{(k)} \in \arg\min_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{w}^{(k)})$.
2. Evaluate $\theta(\boldsymbol{w}^{(k)}) = L(\boldsymbol{x}^{(k)}, \boldsymbol{w}^{(k)})$ and $\boldsymbol{\xi}^{(k)} = \boldsymbol{r}(\boldsymbol{x}^{(k)})$.
3. Record the best dual point so far and update weighted sums of the evaluated dual and primal points.
4. Set $\boldsymbol{w}^{(k+1)} = \Pi_W(\boldsymbol{w}^{(k)} + \alpha^{(k)}\boldsymbol{\xi}^{(k)})$.

Return the best evaluated multiplier and the weighted averages, together with primal feasibility diagnostics. A small step alone is not a stopping certificate.

The best evaluated value $\theta^{(K)}_{\text{best}} = \max_{0\le k<K} \theta(\boldsymbol{w}^{(k)})$ is a valid lower bound on the primal optimum. If a primal feasible point $\widehat{\boldsymbol{x}}$ is available, the computable gap

$$f(\widehat{\boldsymbol{x}}) - \theta^{(K)}_{\text{best}} \le \varepsilon_{\text{out}}$$

certifies the corresponding objective accuracy. An infeasible primal average cannot be used as the upper bound without further justification. With a diminishing step sequence, small multiplier displacements may result from the coefficient alone, even far from optimality. They are not a substitute for this gap.

20.4.3 Diminishing steps and convergence

For $q(w) = -|w|$, start at $w^{(0)} = \alpha/2$ with constant $\alpha > 0$. The unique supergradient away from zero is $-\operatorname{sign}(w)$, so

$$\alpha/2 \;\longmapsto\; -\alpha/2 \;\longmapsto\; \alpha/2 \;\longmapsto \cdots .$$

The function values stay at $-\alpha/2$, below the optimum zero. Making the step small reduces the oscillation but does not remove it.

Decreasing the step too rapidly creates the opposite problem. Start from $w^{(0)} = 2$ and take $\alpha^{(k)} = 1/[2(k+1)^2]$. The total movement is $\sum_k \alpha^{(k)} = \pi^2/12 < 2$, so the iterates remain positive and converge to $2 - \pi^2/12 > 0$, never reaching the optimizer. These examples motivate two requirements:

$$\alpha^{(k)} \longrightarrow 0 \qquad \text{and} \qquad \sum_{k=0}^{\infty} \alpha^{(k)} = \infty.$$

The first lets oscillations shrink; the second leaves enough total movement. They are a broadly useful guarantee for the basic nonsmooth method, not necessary conditions for convergence on every particular problem. Smooth problems and special step rules can behave differently.

20.4.4 What ergodic convergence means

We will analyze K evaluated points indexed $0, \dots, K-1$. Define

$$S_K = \sum_{k=0}^{K-1} \alpha^{(k)}, \qquad \overline{\boldsymbol{w}}^{(K)} = \frac{1}{S_K} \sum_{k=0}^{K-1} \alpha^{(k)} \boldsymbol{w}^{(k)}, \qquad \theta_{\text{best}}^{(K)} = \max_{0 \le k < K} \theta(\boldsymbol{w}^{(k)}). \tag{20.17}$$

The term *ergodic* refers here to the weighted average $\overline{\boldsymbol{w}}^{(K)}$, rather than to the last point. Convexity of C keeps this average feasible, and concavity gives

$$\theta(\overline{\boldsymbol{w}}^{(K)}) \ge \frac{1}{S_K} \sum_{k=0}^{K-1} \alpha^{(k)} \theta(\boldsymbol{w}^{(k)}).$$

There need not be a fixed ordering between the value at the averaged point and the best individual value. Both, however, will have the same upper bound on their objective error.

Theorem 20.6 (Ergodic and best-value dual bounds)**.** *Let C be nonempty, closed, and convex. Suppose θ is concave and finite on C, an optimizer $\boldsymbol{w}^* \in C$ attains $d^* = \theta(\boldsymbol{w}^*)$, and the projected iteration uses exact supergradients satisfying $\left\|\boldsymbol{\xi}^{(k)}\right\| \le G$. Put $R = \left\|\boldsymbol{w}^{(0)} - \boldsymbol{w}^*\right\|$. Then*

$$\max\{d^* - \theta_{\text{best}}^{(K)},\, d^* - \theta(\overline{\boldsymbol{w}}^{(K)})\} \le B_K \coloneqq \frac{R^2 + G^2 \sum_{k=0}^{K-1} (\alpha^{(k)})^2}{2 S_K}. \tag{20.18}$$

Proof. Let $\Delta_k = d^* - \theta(\boldsymbol{w}^{(k)}) \ge 0$. Since $\Pi_C(\boldsymbol{w}^*) = \boldsymbol{w}^*$, nonexpansiveness gives

$$\begin{aligned} \left\|\boldsymbol{w}^{(k+1)} - \boldsymbol{w}^*\right\|^2 &\le \left\|\boldsymbol{w}^{(k)} + \alpha^{(k)} \boldsymbol{\xi}^{(k)} - \boldsymbol{w}^*\right\|^2 \\ &= \left\|\boldsymbol{w}^{(k)} - \boldsymbol{w}^*\right\|^2 + 2\alpha^{(k)} (\boldsymbol{\xi}^{(k)})^{\mathsf{T}} (\boldsymbol{w}^{(k)} - \boldsymbol{w}^*) + (\alpha^{(k)})^2 \left\|\boldsymbol{\xi}^{(k)}\right\|^2 . \end{aligned}$$

The supporting inequality at $\boldsymbol{w}^{(k)}$, evaluated at $\boldsymbol{w}^*$, states $(\boldsymbol{\xi}^{(k)})^\mathsf{T}(\boldsymbol{w}^{(k)} - \boldsymbol{w}^*) \le -\Delta_k$. Therefore

$$\left\|\boldsymbol{w}^{(k+1)} - \boldsymbol{w}^*\right\|^2 \le \left\|\boldsymbol{w}^{(k)} - \boldsymbol{w}^*\right\|^2 - 2\alpha^{(k)}\Delta_k + G^2(\alpha^{(k)})^2. \tag{20.19}$$

Rearranging and summing from $k = 0$ to $K - 1$ telescopes the distances:

$$2\sum_{k=0}^{K-1} \alpha^{(k)}\Delta_k \le R^2 - \left\|\boldsymbol{w}^{(K)} - \boldsymbol{w}^*\right\|^2 + G^2 \sum_{k=0}^{K-1} (\alpha^{(k)})^2.$$

Discard the nonpositive final-distance term and divide by $2S_K$. The minimum of the gaps is no larger than their weighted average, giving the best-value bound. Concavity at $\overline{\boldsymbol{w}}^{(K)}$ gives the same bound for its gap. □

The assumptions describe the selected supergradients along the actual run; θ need not be globally Lipschitz on the entire multiplier space. A bounded set of inner solutions and continuous residuals supplies such a bound. A local Danskin assumption at one point alone does not.

Theorem 20.7 (Nonsummable diminishing steps)**.** *Under the assumptions of Theorem 20.6, suppose*

$$\alpha^{(k)} > 0, \qquad \alpha^{(k)} \to 0, \qquad S_K \to \infty.$$

Then $\theta^{(K)}_{\text{best}} \to d^*$ *and* $\theta(\overline{\boldsymbol{w}}^{(K)}) \to d^*$.

Proof. It remains to show $\sum_{k<K}(\alpha^{(k)})^2/S_K \to 0$. Given $\eta > 0$, choose N so that $\alpha^{(k)} \le \eta$ for every $k \ge N$. For $K > N$,

$$\frac{\sum_{k=0}^{K-1}(\alpha^{(k)})^2}{S_K} \le \frac{\sum_{k=0}^{N-1}(\alpha^{(k)})^2}{S_K} + \eta\frac{\sum_{k=N}^{K-1}\alpha^{(k)}}{S_K} \le \frac{\sum_{k=0}^{N-1}(\alpha^{(k)})^2}{S_K} + \eta.$$

The finite-prefix term tends to zero. Taking the upper limit and then letting $\eta \downarrow 0$ proves the ratio limit. Also $R^2/S_K \to 0$, so Equation (20.18) tends to zero. □

This proof concerns function values at the best and averaged points. It does not, by itself, prove convergence of the raw multiplier sequence or pointwise primal feasibility.

20.4.5 A stronger condition for convergence of the iterates

The frequently used conditions

$$\sum_{k=0}^{\infty} \alpha^{(k)} = \infty, \qquad \sum_{k=0}^{\infty} (\alpha^{(k)})^2 < \infty \tag{20.20}$$

imply diminishing steps. Their additional strength controls the accumulated positive error term in the distance recursion.

Theorem 20.8 (Convergence with square-summable steps)**.** *Under the assumptions of Theorem 20.6, assume Equation (20.20) and that θ is upper semicontinuous on C. Then $\boldsymbol{w}^{(k)}$ converges to a dual optimizer. For a dual function defined as an infimum of continuous affine functions of $\boldsymbol{w}$, upper semicontinuity holds automatically.*

Proof. For any optimizer $\boldsymbol{w}^*$, define

$$a_k = \left\|\boldsymbol{w}^{(k)} - \boldsymbol{w}^*\right\|^2, \qquad e_k = G^2(\alpha^{(k)})^2.$$

By Equation (20.19), $a_{k+1} \le a_k + e_k$. Consequently $a_k + \sum_{j=k}^{\infty} e_j$ is nonincreasing and bounded below. It converges, and its tail sum tends to zero, so a_k has a finite limit. In particular, the iterates are bounded. Summing the full distance recursion also gives $\sum_k \alpha^{(k)}\Delta_k < \infty$. Since $\sum_k \alpha^{(k)} = \infty$, a subsequence satisfies $\Delta_{k_j} \to 0$. Boundedness supplies a further subsequence converging to some $\widehat{\boldsymbol{w}} \in C$. Upper semicontinuity gives

$$\theta(\widehat{\boldsymbol{w}}) \ge \limsup_j \theta(\boldsymbol{w}^{(k_j)}) = d^*.$$

Hence $\widehat{\boldsymbol{w}}$ is optimal. The first part of the proof applies with $\boldsymbol{w}^* = \widehat{\boldsymbol{w}}$, so $\left\|\boldsymbol{w}^{(k)} - \widehat{\boldsymbol{w}}\right\|$ has a limit. Its convergent subsequence has limit zero, forcing the whole sequence to converge to $\widehat{\boldsymbol{w}}$. Finally, the infimum of continuous affine functions is upper semicontinuous because its strict sublevel sets are unions of open sets. □

20.4.6 Rates and the connection to usual subgradient steps

For a constant $\alpha^{(k)} = \alpha > 0$, the bound becomes

$$B_K = \frac{R^2}{2\alpha K} + \frac{\alpha G^2}{2}. \tag{20.21}$$

The first term vanishes, but the second remains. This is a worst-case error bound, not a claim that every problem must have a positive limiting error.

If a finite horizon K is fixed in advance and $R, G > 0$ are known bounds, choose the constant coefficient for that run as

$$\alpha^{(k)} = \frac{R}{G\sqrt{K}} \quad (0 \le k < K).$$

Substituting gives $B_K = RG/\sqrt{K}$. Thus $O(\varepsilon_{\text{out}}^{-2})$ oracle evaluations suffice for this generic objective guarantee. The coefficient depends on the planned horizon; it is not a claim about continuing one positive constant step forever.

For an anytime schedule, take $\alpha^{(k)} = a/(k+1)^q$ with $a > 0$. Integral comparisons of sums yield the following consequences of the same weighted-average bound:

Exponent	Conditions	Bound B_K
$0 < q < 1/2$	diminishing, nonsummable	$O(K^{-q})$
$q = 1/2$	diminishing, nonsummable	$O(\log K/\sqrt{K})$
$1/2 < q < 1$	also square-summable	$O(K^{q-1})$
$q = 1$	also square-summable	$O(1/\log K)$

For example, when $q = 1/2$, S_K is of order $\sqrt{K}$ and $\sum_{k<K}(\alpha^{(k)})^2$ is of order $\log K$. Although the latter sum diverges, dividing by the cumulative step weight gives

$$\frac{\sum_{k<K}(\alpha^{(k)})^2}{S_K} = O\left(\frac{\log K}{\sqrt{K}}\right) \longrightarrow 0.$$

Thus this rule still yields convergence of the best and weighted-average objective values. The accumulated second-order error need not stay finite; it must become negligible relative to S_K.

This argument does not itself prove convergence of the last objective value or of the iterates. When $q > 1/2$ the square sum is bounded, while S_K is of order K^{1-q} for $q < 1$ and $\log K$ for $q = 1$. These are upper bounds from a general analysis; an individual example can converge faster.

There is no different step-size theory caused by the word "dual." If $F = -\theta$ and $\boldsymbol{s}^{(k)} = -\boldsymbol{\xi}^{(k)} \in \partial F(\boldsymbol{w}^{(k)})$, then the very same update is

$$\boldsymbol{w}^{(k+1)} = \Pi_C\big(\boldsymbol{w}^{(k)} - \alpha^{(k)}\boldsymbol{s}^{(k)}\big).$$

This is ordinary projected convex subgradient descent. Apparent differences between step prescriptions usually concern the output being averaged, a finite-horizon versus asymptotic guarantee, or stochastic error.

For example, a common $1/\sqrt{k+1}$ rule also gives an $O(1/\sqrt{K})$ bound for a *uniform* average under a uniform distance bound. To see why this does not contradict the table, assume $\left\|\boldsymbol{w}^{(k)} - \boldsymbol{w}^*\right\| \le D$ for all k and that the steps are nonincreasing. Put $a_k = \left\|\boldsymbol{w}^{(k)} - \boldsymbol{w}^*\right\|^2$ and divide Equation (20.19) by $2\alpha^{(k)}$ before summing. Summation by parts gives

$$\begin{aligned}\sum_{k=0}^{K-1} \Delta_k &\le \frac{1}{2}\sum_{k=0}^{K-1} \frac{a_k - a_{k+1}}{\alpha^{(k)}} + \frac{G^2}{2}\sum_{k=0}^{K-1} \alpha^{(k)} \\ &\le \frac{D^2}{2\alpha^{(K-1)}} + \frac{G^2}{2}\sum_{k=0}^{K-1} \alpha^{(k)}.\end{aligned}$$

With $\alpha^{(k)} = a/\sqrt{k+1}$, division by K and $\sum_{k=0}^{K-1}(k+1)^{-1/2} \le 2\sqrt{K}$ give

$$d^* - \theta\left(\frac{1}{K}\sum_{k=0}^{K-1} \boldsymbol{w}^{(k)}\right) \le \frac{D^2/(2a) + aG^2}{\sqrt{K}}.$$

A bounded set C supplies the distance assumption. This argument uses a different average and a different assumption; the logarithmic factor in the earlier bound should not simply be deleted.

In a stochastic method, the sampled vector is generally not an exact supergradient at every iteration. Under conditional unbiasedness and a uniform conditional second-moment bound, taking conditional expectations in the distance expansion gives an expected version of the ergodic bound. The square-summable, nonsummable conditions are also commonly imposed to control accumulated noise in almost-sure convergence results. Those claims require the stochastic assumptions; a biased inexact Lagrangian solve is not automatically such an oracle. Finally, when the dual has a Lipschitz continuous gradient, sufficiently small constant steps can converge by smooth optimization theory. The generic nonsmooth guarantee does not preclude that special case.

20.4.7 Primal recovery from averages

We now derive feasibility and objective bounds for the primal average in Equation (20.29), using the original projection set $W = \mathbb{R}^m \times \mathbb{R}^p_+$. Extra projection onto a restricted set C changes the multiplier identities used here, so these particular primal bounds should not be transferred to that case without a new argument.

Theorem 20.9 (Ergodic primal recovery)**.** *Suppose f and every h_j are convex, $\boldsymbol{g}$ is affine, a primal optimizer $\boldsymbol{x}^*$ exists, the Lagrangian subproblems are solved globally, and $\left\|\boldsymbol{r}(\boldsymbol{x}^{(k)})\right\| \le G$. Under Equations (20.15) and (20.16),*

$$\boldsymbol{g}(\overline{\boldsymbol{x}}^{(K)}) = \frac{\boldsymbol{\lambda}^{(K)} - \boldsymbol{\lambda}^{(0)}}{S_K}, \tag{20.22}$$

$$h_j(\overline{\boldsymbol{x}}^{(K)}) \le \frac{\nu_j^{(K)} - \nu_j^{(0)}}{S_K} \le \frac{\nu_j^{(K)}}{S_K}, \tag{20.23}$$

$$f(\overline{\boldsymbol{x}}^{(K)}) - p^* \le \frac{\left\|\boldsymbol{w}^{(0)}\right\|^2 + G^2 \sum_{k=0}^{K-1} (\alpha^{(k)})^2}{2S_K}. \tag{20.24}$$

The objective bound is one-sided because the average need not yet be feasible.

Proof. Summing the equality updates gives $\sum_{k<K} \alpha^{(k)} \boldsymbol{g}(\boldsymbol{x}^{(k)}) = \boldsymbol{\lambda}^{(K)} - \boldsymbol{\lambda}^{(0)}$. An affine function preserves convex combinations, proving Equation (20.22). For inequalities, componentwise projection gives

$$\nu_j^{(k+1)} \ge \nu_j^{(k)} + \alpha^{(k)} h_j(\boldsymbol{x}^{(k)}).$$

Summation followed by convexity of h_j gives Equation (20.23); its second inequality uses $\nu_j^{(0)} \ge 0$.

Global inner optimality and primal feasibility imply

$$f(\boldsymbol{x}^{(k)}) + (\boldsymbol{w}^{(k)})^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}^{(k)}) = L(\boldsymbol{x}^{(k)}, \boldsymbol{w}^{(k)}) \le L(\boldsymbol{x}^*, \boldsymbol{w}^{(k)}) \le p^*.$$

Because $\mathbf{0} \in W$, nonexpansiveness also gives

$$\left\|\boldsymbol{w}^{(k+1)}\right\|^2 \le \left\|\boldsymbol{w}^{(k)}\right\|^2 + 2\alpha^{(k)} (\boldsymbol{w}^{(k)})^{\mathsf{T}} \boldsymbol{r}(\boldsymbol{x}^{(k)}) + G^2 (\alpha^{(k)})^2.$$

Combining and summing yields

$$\sum_{k=0}^{K-1} \alpha^{(k)} (f(\boldsymbol{x}^{(k)}) - p^*) \le \frac{\left\|\boldsymbol{w}^{(0)}\right\|^2 - \left\|\boldsymbol{w}^{(K)}\right\|^2 + G^2 \sum_{k=0}^{K-1} (\alpha^{(k)})^2}{2}.$$

Discard the nonpositive final norm, divide by S_K, and use convexity of f. □

If the multipliers are bounded, $S_K \to \infty$, and $\sum_{k<K} (\alpha^{(k)})^2 / S_K \to 0$, the first two bounds imply

$$\left\|\boldsymbol{g}(\overline{\boldsymbol{x}}^{(K)})\right\| \to 0, \tag{20.25}$$

$$\left\|[\boldsymbol{h}(\overline{\boldsymbol{x}}^{(K)})]_+\right\| \to 0. \tag{20.26}$$

Any cluster point of the primal averages is therefore feasible, by continuity. The objective bound and continuity show that its value is at most p^*; feasibility gives the reverse inequality. Thus every such cluster point is primal optimal. This reasoning requires existence of a cluster point; dual convergence alone need not make the primal averages bounded.

A finite dual optimizer with zero gap also gives a direct lower estimate for an infeasible average:

$$f(\overline{\boldsymbol{x}}^{(K)}) - p^* \ge -(\boldsymbol{\lambda}^*)^{\mathsf{T}} \boldsymbol{g}(\overline{\boldsymbol{x}}^{(K)}) - (\boldsymbol{\nu}^*)^{\mathsf{T}} \boldsymbol{h}(\overline{\boldsymbol{x}}^{(K)}) \ge -\left\|\boldsymbol{\lambda}^*\right\| \left\|\boldsymbol{g}(\overline{\boldsymbol{x}}^{(K)})\right\| - \left\|\boldsymbol{\nu}^*\right\| \left\|[\boldsymbol{h}(\overline{\boldsymbol{x}}^{(K)})]_+\right\|.$$

Together with the upper bound, vanishing feasibility violation then forces $f(\overline{\boldsymbol{x}}^{(K)}) \to p^*$, even without assuming that the primal averages themselves converge. Under the dual assumptions used for the main distance bound, bounded multipliers can be replaced by a consequence of that bound. Namely,

$$\left\|\boldsymbol{w}^{(K)} - \boldsymbol{w}^*\right\|^2 \le R^2 + G^2 \sum_{k=0}^{K-1} (\alpha^{(k)})^2.$$

If $S_K \to \infty$ and $\sum_{k<K}(\alpha^{(k)})^2/S_K \to 0$, dividing the square-root bound by S_K shows $\left\|\boldsymbol{w}^{(K)}\right\|/S_K \to 0$. The feasibility identities therefore still tend to zero. Thus, with convex primal data and attained matching primal–dual optima, nonsummable diminishing steps yield convergence of the averaged primal feasibility and objective value under the stated residual bound, without needing convergence of the individual inner solutions. Positive duality gap or nonconvex primal data removes these general recovery guarantees.

20.5 A common numerical test

20.5.1 Constructing the primal problem and the exact oracle

To see the nonsmooth behavior in an explicit constrained problem, consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} f(\boldsymbol{x}) = \frac{1}{2}\big(\max\{|x_1| - 1, 0\}\big)^2 + \frac{1}{2}x_2^2 \quad \text{subject to} \quad x_1 - \frac{1}{4} = 0, \quad 1 - x_2 \le 0. \tag{20.27}$$

The first term is half the squared distance to $[-1, 1]$. Its derivative is $x_1 + 1$ for $x_1 < -1$, zero for $|x_1| \le 1$, and $x_1 - 1$ for $x_1 > 1$. The derivative is continuous and nondecreasing, so this term is convex and continuously differentiable. The equality fixes $x_1 = 1/4$, where the first term vanishes. Minimizing the remaining term $x_2^2/2$ over $x_2 \ge 1$ therefore gives $\boldsymbol{x}^* = (1/4, 1)^\mathsf{T}$ and $p^* = 1/2$. The point $(1/4, 2)^\mathsf{T}$ satisfies Slater's condition.

We derive the inner solution by separating the two coordinates. The second coordinate minimizes $x_2^2/2 - \nu x_2 + \nu$, so $x_2(\nu) = \nu$ and its contribution is $\nu - \nu^2/2$. If $\lambda > 0$, the first-coordinate minimum is to the left of the flat interval and solves $x_1 + 1 + \lambda = 0$. If $\lambda < 0$, it lies to the right and solves $x_1 - 1 + \lambda = 0$. At $\lambda = 0$, every point of $[-1, 1]$ minimizes that term. Thus a convenient oracle is

$$x_1(\lambda) = \begin{cases} -\operatorname{sign}(\lambda)(1 + |\lambda|), & \lambda \ne 0, \\ -1, & \lambda = 0, \end{cases} \qquad x_2(\nu) = \nu.$$

Substitution gives

$$\theta(\lambda, \nu) = -|\lambda| - \tfrac{1}{2}\lambda^2 - \tfrac{1}{4}\lambda + \nu - \tfrac{1}{2}\nu^2, \qquad \nu \ge 0. \tag{20.28}$$

We can maximize the two terms separately: $\nu - \nu^2/2$ is maximized at $\nu = 1$, while $-|\lambda| - \lambda^2/2 - \lambda/4$ is strictly increasing for $\lambda < 0$ and strictly decreasing for $\lambda > 0$. Thus the maximum is attained at $(\lambda^*, \nu^*) = (0, 1)$, with $d^* = 1/2$. At that point

$$X(0, 1) = [-1, 1] \times \{1\}, \qquad \partial^+\theta(0, 1) = [-5/4, 3/4] \times \{0\}.$$

The inner solution is not unique and its residuals differ, so the dual has a corner. Zero belongs to the superdifferential, but the chosen tie-breaking rule returns the nonzero residual $(-5/4, 0)^\mathsf{T}$. Thus the method can leave an optimal multiplier even with an exact inner solve.

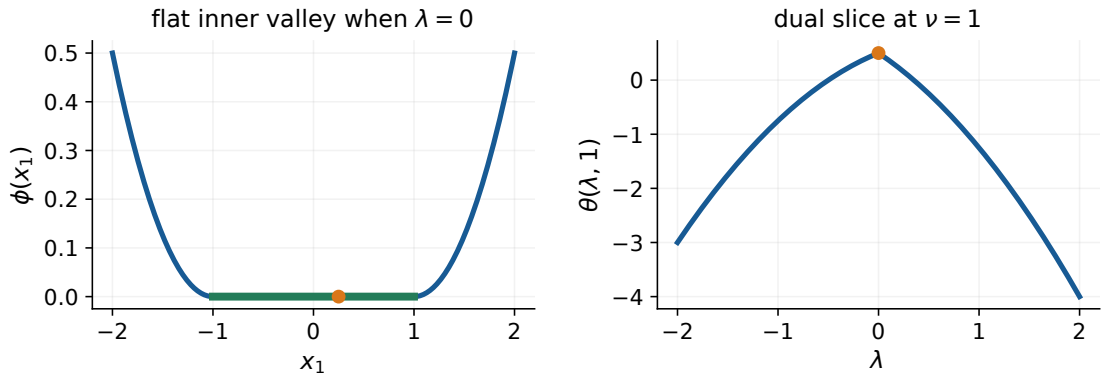


Figure 20.4: The common test has a continuously differentiable primal objective with a flat interval in its first coordinate. Minimizing over that coordinate produces the corner in the dual slice at $\nu = 1$.

20.5.2 Comparing prescribed step sequences

The code file is `l20_dual_subgradient.py`. Its function `projected_supergradient` applies the basic update using an oracle, a projection, and a prescribed step rule. The exact inner solution is supplied by `lagrangian_oracle`. The function `run_comparison` compares schedules on this common problem. The common start is $(\lambda^{(0)}, \nu^{(0)}) = (1, 0)$, and each run makes $K = 2000$ oracle evaluations. The schedules are a constant 0.2 and $0.5/(k+1)^q$ for $q = 1/2, 3/4, 1$. The optimal value is used only to report errors; it is not used to choose a step. The last reported point is $\boldsymbol{w}^{(K-1)}$, and both averages use the evaluated points with indices $0, \dots, K-1$.

Before comparing the output, we can check the assumptions of the convergence bound on this example. All inner minima are attained throughout $W = \mathbb{R} \times \mathbb{R}_+$. Each implemented step lies in $(0, 1]$. Its equality multiplier update is a convex combination of the current λ and $-5/4$ when $\lambda \geq 0$, or of the current λ and $3/4$ when $\lambda < 0$. Thus $\lambda^{(k)} \in [-5/4, 1]$ for this start. Similarly, $\nu^{(k)} \in [0, 1]$. It follows that

$$|g(\boldsymbol{x}^{(k)})| \leq 9/4, \qquad |h(\boldsymbol{x}^{(k)})| \leq 1, \qquad \left\|\boldsymbol{r}(\boldsymbol{x}^{(k)})\right\| \leq \sqrt{97}/4.$$

The selected supergradients are therefore uniformly bounded along all four runs.

The following results distinguish the raw last-point error, the best error, and the value at the weighted dual average. The averaged dual gap is $d^* - \theta(\overline{\boldsymbol{w}}^{(K)})$, not the average of the individual gaps.

Step schedule	Last dual gap	Best dual gap	Averaged dual gap
0.2	3.8389×10^{-2}	2.0376×10^{-2}	3.7902×10^{-2}
$0.5/(k+1)^{1/2}$	2.5230×10^{-3}	3.4976×10^{-6}	1.9500×10^{-3}
$0.5/(k+1)^{3/4}$	5.1405×10^{-4}	7.6556×10^{-8}	5.0039×10^{-2}
$0.5/(k+1)$	1.9598×10^{-4}	1.0229×10^{-4}	1.7488×10^{-1}

The table compares outputs at the same finite budget. A quickly decaying step can give an excellent current point while its weighted average still retains substantial influence from early points. This explains the different rankings of the last and averaged columns; it does not imply an asymptotic failure of averaging. If $\boldsymbol{w}^{(k)} \to \boldsymbol{w}^*$ and $S_K \to \infty$, split the average into a finite prefix and a tail. The prefix weight divided by S_K tends to zero, and the tail lies arbitrarily close to $\boldsymbol{w}^*$, so $\overline{\boldsymbol{w}}^{(K)} \to \boldsymbol{w}^*$ as well.

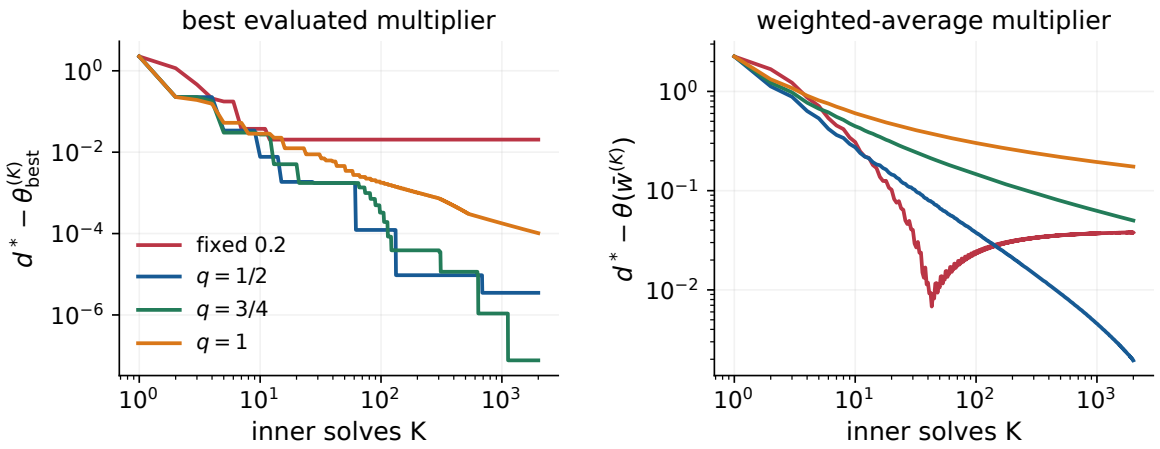


Figure 20.5: Dual objective progress on the common test. Raw values can oscillate, while best values and averages measure different aspects of a run.

20.5.3 Dual iterates and the associated primal points

We now follow the $q = 1/2$ run to distinguish multiplier convergence from primal recovery. Write $a_k = 0.5/\sqrt{k+1}$ for its step coefficient. The explicit oracle gives

$$\nu^{(k+1)} = (1 - a_k)\nu^{(k)} + a_k,$$
$$\lambda^{(k+1)} = \begin{cases} (1 - a_k)\lambda^{(k)} - a_k(\operatorname{sign}(\lambda^{(k)}) + 1/4), & \lambda^{(k)} \neq 0, \\ -5a_k/4, & \lambda^{(k)} = 0. \end{cases}$$

The projection in the first line is inactive because $\nu^{(k)} \in [0, 1]$. Subtracting that line from one and iterating gives

$$1 - \nu^{(k)} = \prod_{j=0}^{k-1}(1 - a_j) \leq \exp\left(-\sum_{j=0}^{k-1} a_j\right) \longrightarrow 0.$$

For the equality multiplier, as long as a step preserves its nonzero sign, its magnitude decreases by at least $3a_k/4$. It cannot keep a fixed nonzero sign indefinitely: since $\sum_k a_k = \infty$, summing those decreases would eventually make its magnitude negative. When a step crosses zero, its overshoot has magnitude at most $5a_k/4$; the same bound holds for a step starting at zero. Since $a_k \to 0$, all sufficiently late overshoots are arbitrarily small, and magnitudes decrease between crossings. Hence $\lambda^{(k)} \to 0$ and $\boldsymbol{w}^{(k)} \to (0, 1)$ for this particular problem and schedule. This conclusion uses the explicit recurrence, rather than treating the ergodic objective theorem as a theorem about every raw iterate.

At $K = 2000$, the last evaluated multiplier is approximately $(-0.00335645, 1)$ and the weighted average is $(0.00135262, 0.97727106)$. Their distances to $\boldsymbol{w}^*$ are approximately 0.00335645 and 0.02276915, respectively.

To see what this implies for the primal variables, substitute these multipliers back into the oracle. For $\lambda \neq 0$, $x_1(\lambda) = -\operatorname{sign}(\lambda) - \lambda$; at zero the chosen value is -1. Thus every returned x_1 lies outside $(-1, 1)$ and

$$|g(\boldsymbol{x}^{(k)})| = |x_1^{(k)} - 1/4| \geq 3/4.$$

The raw primal points therefore cannot become feasible, even though the multipliers converge. The sign changes visible in Figure 20.6 select opposite sides of the flat inner valley, which

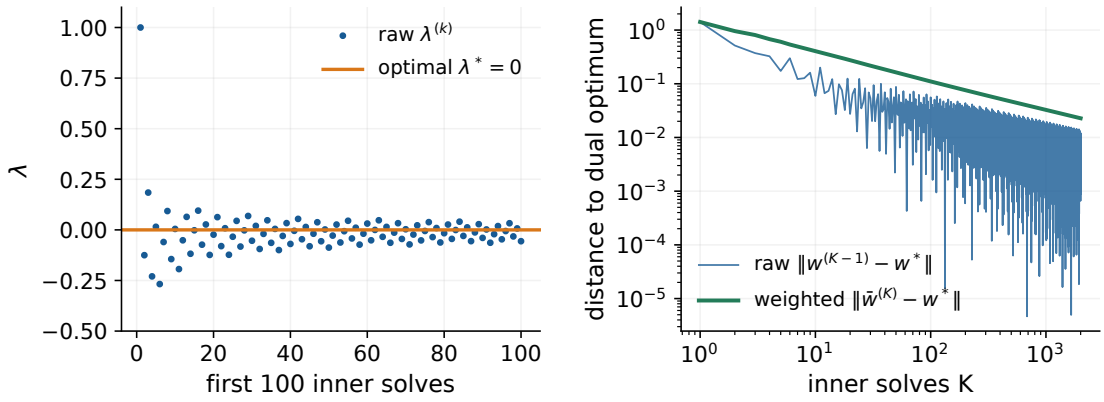


Figure 20.6: The $q = 1/2$ run: equality-multiplier oscillations shrink, and both evaluated multipliers and their weighted averages approach $\boldsymbol{w}^* = (0, 1)$. The right plot uses $\boldsymbol{w}^{(K-1)}$ after K inner solves.

explains the persistent switching in Figure 20.7. To combine information from those different inner solutions, form the step-weighted primal average

$$\overline{\boldsymbol{x}}^{(K)} = \frac{1}{S_K} \sum_{k=0}^{K-1} \alpha^{(k)} \boldsymbol{x}^{(k)}. \tag{20.29}$$

This is an average of primal points, distinct from the multiplier average in Equation (20.17).

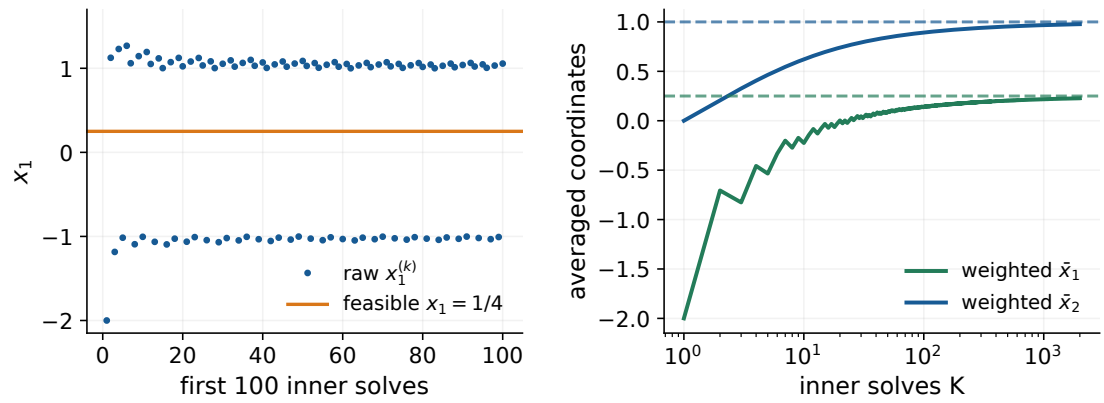


Figure 20.7: For the same run as Figure 20.6, raw inner solutions keep switching across $x_1 = 1/4$, while the weighted primal coordinates approach the feasible values.

For completeness, the equality and positive-part inequality residuals of the weighted primal averages are, respectively, $(3.0643 \times 10^{-3}, 2.5000 \times 10^{-3})$, $(2.2614 \times 10^{-2}, 2.2729 \times 10^{-2})$, $(8.5751 \times 10^{-2}, 8.5800 \times 10^{-2})$, and $(2.4454 \times 10^{-1}, 2.4146 \times 10^{-1})$ in the table's order. A small finite-run primal violation with constant steps does not remove their generic dual oscillation. To determine when averaging has a theoretical recovery guarantee, we next use the multiplier updates themselves.

20.6 Implementation details and comparison with ALM

20.6.1 Inexact inner solves

Suppose a global inner objective-error bound is available:

$$0 \le L(\widetilde{\boldsymbol{x}}^{(k)}, \boldsymbol{w}^{(k)}) - \theta(\boldsymbol{w}^{(k)}) \le \varepsilon_{\text{in}}^{(k)}. \tag{20.30}$$

For any multiplier $\boldsymbol{z}$, evaluate its Lagrangian at this approximate point:

$$\begin{aligned}\theta(\boldsymbol{z}) &\le L(\widetilde{\boldsymbol{x}}^{(k)}, \boldsymbol{z}) \\ &= L(\widetilde{\boldsymbol{x}}^{(k)}, \boldsymbol{w}^{(k)}) + \boldsymbol{r}(\widetilde{\boldsymbol{x}}^{(k)})^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{w}^{(k)}) \\ &\le \theta(\boldsymbol{w}^{(k)}) + \varepsilon_{\text{in}}^{(k)} + \boldsymbol{r}(\widetilde{\boldsymbol{x}}^{(k)})^{\mathsf{T}}(\boldsymbol{z} - \boldsymbol{w}^{(k)}).\end{aligned}$$

Thus the residual is an $\varepsilon_{\text{in}}^{(k)}$-supergradient. Evaluating this inequality at an optimum changes the distance recursion by adding $2\alpha^{(k)}\varepsilon_{\text{in}}^{(k)}$. Telescoping exactly as before gives

$$\max\{d^* - \theta_{\text{best}}^{(K)},\ d^* - \theta(\overline{\boldsymbol{w}}^{(K)})\} \le B_K + \frac{\sum_{k=0}^{K-1} \alpha^{(k)}\varepsilon_{\text{in}}^{(k)}}{S_K}, \tag{20.31}$$

provided the selected approximate residuals remain bounded by G. Here $\theta_{\text{best}}^{(K)}$ and $\theta(\overline{\boldsymbol{w}}^{(K)})$ refer to *true* dual values at the generated points. They describe a theoretical error guarantee; approximate inner values $L(\widetilde{\boldsymbol{x}}^{(k)}, \boldsymbol{w}^{(k)})$ cannot simply be substituted for them. We return below to the lower bound that can actually be certified. Consequently a sufficient additional condition for ergodic convergence is that the weighted average of inner errors tends to zero. A fixed positive inner error generally leaves an accuracy floor; summability of $\sum_k \alpha^{(k)}\varepsilon_{\text{in}}^{(k)}$ is a stronger useful condition. The primal objective bound acquires the same weighted error term, since the inner comparison with a feasible $\boldsymbol{x}^*$ now has that extra error.

These are objective-gap assumptions, not merely inner first-order tolerances. They must be justified by a global solver or by suitable convexity and error bounds. If Equation (20.30) is certified, then $L(\widetilde{\boldsymbol{x}}^{(k)}, \boldsymbol{w}^{(k)}) - \varepsilon_{\text{in}}^{(k)}$ is a certified dual lower bound and hence a primal lower bound for $\boldsymbol{w}^{(k)} \in W$. The uncorrected value $L(\widetilde{\boldsymbol{x}}^{(k)}, \boldsymbol{w}^{(k)})$ need not be one.

20.6.2 Separability, scaling, and an optional Polyak step

Suppose the primal variables separate into blocks and

$$f(\boldsymbol{x}) = \sum_{i=1}^{N} f_i(\boldsymbol{x}_i), \qquad \boldsymbol{r}(\boldsymbol{x}) = \sum_{i=1}^{N} \boldsymbol{r}_i(\boldsymbol{x}_i) - \boldsymbol{b}.$$

The separation result from Chapter 18 gives

$$\theta(\boldsymbol{w}) = -\boldsymbol{w}^{\mathsf{T}}\boldsymbol{b} + \sum_{i=1}^{N} \inf_{\boldsymbol{x}_i \in \mathbb{R}^{n_i}} \{f_i(\boldsymbol{x}_i) + \boldsymbol{w}^{\mathsf{T}}\boldsymbol{r}_i(\boldsymbol{x}_i)\}. \tag{20.32}$$

To verify the equality directly, every sum is at least the sum of its infima; conversely, independently chosen near-minimizers approach each finite block infimum. If a block infimum is $-\infty$, keep finite points in all other blocks and send that block's value downward. When all block minima are attained, their concatenation is a global inner minimizer and its residual is $\sum_i \boldsymbol{r}_i(\boldsymbol{x}_i) - \boldsymbol{b}$. Independent block solves can therefore be coordinated by one multiplier update. Coupling terms that remain in the Lagrangian prevent this simple decomposition.

Constraint scaling changes these residuals and therefore the multiplier steps. A fixed numerical coefficient has a different effect after a constraint is rescaled. Scaling should be chosen consistently before comparing algorithms; changing it adaptively requires additional analysis.

When d^* is known, an optional Polyak coefficient is

$$\alpha^{(k)} = \gamma \frac{d^* - \theta(\boldsymbol{w}^{(k)})}{\left\|\boldsymbol{\xi}^{(k)}\right\|^2}, \qquad 0 < \gamma < 2.$$

At a nonoptimal point the supergradient cannot be zero. Substitution into Equation (20.19), using the exact norm instead of its bound, gives

$$\left\|\boldsymbol{w}^{(k+1)} - \boldsymbol{w}^*\right\|^2 \leq \left\|\boldsymbol{w}^{(k)} - \boldsymbol{w}^*\right\|^2 - \gamma(2-\gamma)\frac{(d^* - \theta(\boldsymbol{w}^{(k)}))^2}{\left\|\boldsymbol{\xi}^{(k)}\right\|^2}. \tag{20.33}$$

This proves a distance decrease without requiring the dual objective to increase. An unknown or incorrectly estimated optimal value does not justify this exact formula. The basic code comparison uses predetermined steps and does not rely on knowing d^*.

20.6.3 Explicit supergradient steps versus the ALM proximal step

For equality constraints, the ordinary dual method solves

$$\boldsymbol{x}^{(k)} \in \arg\min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}), \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \alpha^{(k)} \boldsymbol{g}(\boldsymbol{x}^{(k)}).$$

The residual is evaluated at a minimizer for the *current* multiplier. ALM instead uses the augmented objective from Lecture 15,

$$A_\rho(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}) = L(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}) + \frac{\rho}{2}\left\|\boldsymbol{g}(\boldsymbol{x})\right\|^2,$$

and updates with the returned augmented minimizer:

$$\boldsymbol{x}^{(k+1)} \in \arg\min_{\boldsymbol{x}} A_\rho(\boldsymbol{x}, \boldsymbol{\lambda}^{(k)}), \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x}^{(k+1)}).$$

As derived in Chapter 19, under the convex equality assumptions and exact attained inner minimization this is equivalent to

$$\boldsymbol{\lambda}^{(k+1)} = \arg\max_{\boldsymbol{z}} \left\{ \theta(\boldsymbol{z}) - \frac{1}{2\rho}\left\|\boldsymbol{z} - \boldsymbol{\lambda}^{(k)}\right\|^2 \right\}.$$

The returned residual is a supergradient at the *new* multiplier, so the relation is implicit. This is the source of the damping: the next multiplier must solve a regularized maximization rather than take an unrestricted explicit step using an old supporting slope. The associated envelope

$$M_\rho \theta(\boldsymbol{\lambda}) = \max_{\boldsymbol{z}} \left\{ \theta(\boldsymbol{z}) - \frac{1}{2\rho}\left\|\boldsymbol{z} - \boldsymbol{\lambda}\right\|^2 \right\}$$

is smooth under the conditions established in Lecture 19, with gradient equal to the ALM residual. Thus ALM can be viewed as optimizing a smooth envelope, whereas the method in this chapter works directly with the nonsmooth dual.

The augmented inner problem may be more costly or less separable, and a large ρ can worsen its conditioning. A fixed ALM parameter is therefore not a contradiction to the diminishing explicit steps discussed here: the two methods solve different inner problems and

use first-order information at different multiplier points. Neither method makes every primal inner point feasible. The advantage of the basic supergradient method is its simple, often decomposable oracle. The general finite-horizon analysis gives an $O(K^{-1/2})$ objective-error bound, or $O(\varepsilon_{\text{out}}^{-2})$ inner oracle evaluations for that guarantee. This statement refers to the horizon-dependent step choice; the $q = 1/2$ anytime rule has the weighted-average bound $O(\log K/\sqrt{K})$ derived earlier. Neither is a local Newton rate.

Bundle methods retain several supporting planes, while smoothing and proximal methods modify the problem or the update to use more information. These are useful extensions, but their guarantees require their own assumptions. The course implementation retains the basic projected method and its step-size comparison.

To place this method among the other constrained algorithms, compare what each outer iteration asks the inner solver to do. QP minimizes a quadratic-penalty objective, usually with increasing ρ and the associated conditioning cost. LN and SQP solve a KKT linear system or a quadratic program, so their main concerns include well-posedness and globalization. Barrier methods maintain a strict interior while reducing the barrier parameter. The dual method instead asks for a global Lagrangian minimum and moves the multipliers using its residual. This is attractive when the inner problem separates, but finite inner values, nonsmooth progress, and primal recovery remain essential checks. No choice of dual algorithm can remove a positive optimal duality gap.

Chapter summary

- A supergradient supports a concave function from above. For the dual, any global Lagrangian minimizer provides its constraint residual as one such vector.
- Under local Danskin assumptions, all supergradients form the convex hull of active residuals. Differentiability means those residuals coincide; a unique inner minimizer is sufficient, but not necessary.
- The normalized minimum-norm supergradient is steepest in the unconstrained local geometry. Computing one residual is usually cheaper, although it need not point uphill.
- Projection enforces multiplier feasibility. Valid inner minimization, effective-domain membership, and bounded residuals remain separate requirements.
- Nonsummable diminishing steps give best-value and weighted-average convergence. Square summability provides a stronger iterate-convergence result; horizon-dependent and anytime rate statements must be distinguished.
- Dual supergradient ascent is subgradient descent on $-\theta$. Primal recovery requires additional convexity, boundedness, and feasibility checks; inexact inner errors must also be controlled.

Further reading. The concave statements here are sign-reversed versions of the convex results in Boyd, Duchi, Pilanci, and Vandenberghe's *Subgradients* and Boyd and Jaehyun Park's *Subgradient Methods* (Stanford EE364b). For directional derivatives and Danskin's theorem, see Dimitri P. Bertsekas, *Convex Optimization Theory*, Section 5.4. For the noise assumptions, see Boyd, Mutapcic, and Duchi's *Stochastic Subgradient Methods*. The preceding derivations also adapt the supergradient and steepest-ascent material in the original MA3236 Lectures 19–20; Frank–Wolfe is outside the scope of the present lecture.

20.7 Exercises

Exercise 20.1 (Supporting slopes and steepest ascent)**.** Let $q(w_1, w_2) = \min\{2w_1 + w_2,\ 2w_1 - 3w_2\}$. Compute $\partial^+ q(\mathbf{0})$ and its minimum-norm point. Find a direction of steepest ascent and determine whether each of the two active slopes is itself an ascent direction.

Exercise 20.2 (Residual update)**.** For one equality and one inequality, write the projected multiplier update when the current residual is $(g, h) = (2, -3)$, the current inequality multiplier is 1, and the step coefficient is $1/2$.

Exercise 20.3 (Projected inequality iteration)**.** Derive and run four iterations of dual supergradient ascent for

$$\min_{x \in \mathbb{R}} \tfrac{1}{2}(x-3)^2 \qquad \text{subject to} \qquad x \le 0$$

using $\alpha^{(k)} = 1/4$ and $\nu^{(0)} = 0$. Report the inner minimizer, residual, multiplier, and dual value at each evaluated point. Explain why a constant step can converge in this example.

Exercise 20.4 (Step-size guarantees)**.** For $\alpha^{(k)} = a/(k+1)^q$, determine which $q > 0$ satisfy nonsummable diminishing steps and which also satisfy square summability. Derive the weighted-average bound's orders for $q = 1/2$, $q = 3/4$, and $q = 1$. Explain why these statements do not conflict with a finite-horizon constant coefficient of order $1/\sqrt{K}$.

Exercise 20.5 (A smooth dual with nonunique inner minimizers)**.** For $f(x_1, x_2) = \tfrac{1}{2}x_1^2 + (x_2^2 - 1)^2$ and the equality $x_1 = 0$, derive the dual function, every inner minimizer, and their residuals. Explain why nonuniqueness does not create a corner in this case.